Instantaneous rate of change of y with respect to x is defined as the limiting case of ratio of the change in y to the change in x.

i.e. $\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$

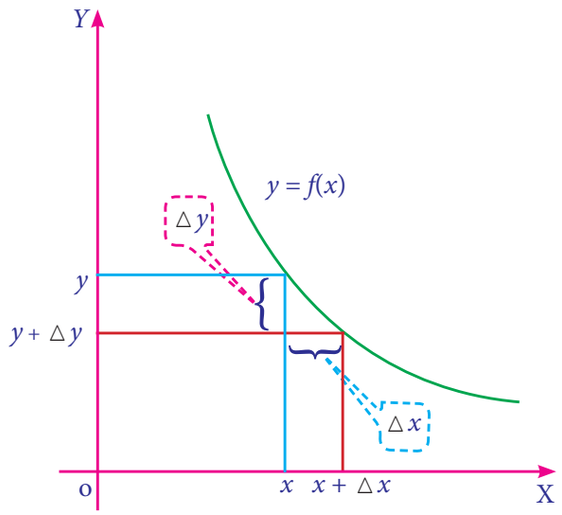


Fig : 6.4(a)

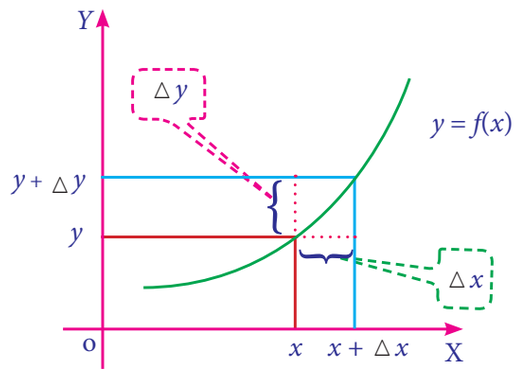


Fig : 6.4(b)

Cost function

The amount spent for the production of a commodity is called its cost function. Normally, total cost function [TC] consists of two parts.

(i) Variable cost; (ii) Fixed cost

Variable cost

Variable cost is the cost which varies almost in direct proportion to the volume of production.

Fixed cost

Fixed cost is the cost which does not vary directly with the volume of production.

If $f(x)$ be the variable cost and k be the fixed cost for production of x units, then total cost is $C(x) = f(x) + k$, $x > 0$.

NOTE

(i) Variable cost $f(x)$ is a single valued function.

(ii) Fixed cost k is independent of the level of output.

(iii) $f(x)$ does not contain constant term.

Some standard results

If $C(x) = f(x) + k$ be the total cost function, then

(i) Average cost:

$$AC = \frac{Total\ Cost}{Output} = \frac{C(x)}{x} = \frac{f(x)+k}{x}$$

(ii) Average variable cost:

$$AVC = \frac{Variable\ Cost}{Output} = \frac{f(x)}{x}$$

(iii) Average fixed cost:

$$AFC = \frac{Fixed\ Cost}{Output} = \frac{k}{x}$$

(iv) Marginal cost:

$$MC = \frac{dC}{dx} = \frac{d}{dx}[C(x)] = C'(x)$$

(v) Marginal average cost:

$$MAC = \frac{d}{dx}(AC)$$

(vi) Total cost:

TC = Average cost × output

(vi) Average cost [AC] is minimum, when $MC = AC$

Remark:

The marginal cost [MC] is approximately equal to the additional production cost of $(x+1)^{th}$ unit, when the production level is x units.

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